



MT1004 – Linear Algebra
Homework # 1

Question: Resource Allocation in Cloud Computing

Problem: Virtual Machine Resource Allocation

A cloud data center provides three types of virtual machines (VMs):

- **VM Type A:** Small
- **VM Type B:** Medium
- **VM Type C:** Large

Each type of VM consumes CPU, memory, and storage resources as shown below.

	A	B	C
CPU Units	2	4	6
Memory Unit	1	3	4
Storage Unit	1	2	5

Suppose that the cloud data center has exactly:

- 40 CPU units
- 25 memory units
- 30 storage units

Assume that all the available resources must be fully utilized.

Let

x = number of Type A VMs

y = number of Type B VMs

z = number of Type C VMs.

How many VMs of each type can be deployed while using these resources?

Tasks

- Formulate a system of three linear equations representing the CPU, memory, and storage constraints.
- Write the system in the matrix form

$$A\mathbf{x} = \mathbf{b}$$

Clearly identify the coefficient matrix A , the unknown vector $\mathbf{x}$, and the resource vector $\mathbf{b}$.

- Construct the augmented matrix corresponding to the system.

- d) Use elementary row operations to convert the augmented matrix into **Row Echelon Form (REF)**.
- e) Hence, determine the number of Type A, Type B, and Type C virtual machines that can be deployed.
- f) Verify your answer by substituting the values of x , y , and z into the original system.
- g) Use Python to verify your solution

```
import numpy as np

# Coefficient matrix
A = np.array([
    [2, 4, 6],
    [1, 3, 4],
    [1, 2, 5]
], dtype=float)

# Resource vector
b = np.array([40, 25, 30], dtype=float)

# Solve the system Ax = b
solution = np.linalg.solve(A, b)

print("Solution:")
print("Type A VMs =", solution[0])
print("Type B VMs =", solution[1])
print("Type C VMs =", solution[2])
```